Analyzing Air Quality and Income Disparities in San Bernardino County, California

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Research question

How do PM2.5 levels compare with the percentage of low-income residents across census block groups in San Bernardino County?

Loading Packages

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(sf)
```

```
Linking to GEOS 3.12.1, GDAL 3.8.4, PROJ 9.4.0; sf_use_s2() is TRUE
```

```
library(here)
```

```
here() starts at /home/katefrankland/MEDS/eds223/EDS223-homework1
```

```
library(tmap)

tmap_mode("plot")
```

```
i tmap modes "plot" - "view"
i toggle with `tmap::ttm()`
```

Reading the EJScreen data

```
ejscreen <- sf::st_read(
  here::here(
    "data",
    "ejscreen",
    "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"
  )
)
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/home/katefrankland/MEDS/eds223/EDS223-homework1/data/ejscreen/EJSCREEN_2023_BG_StatePct_
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

Filtering to California

```
california <- ejscreen |>
  dplyr::filter(ST_ABBREV == "CA")
```

Filtering the data to San Bernardino County in California

```r
san_bernardino_bg <- california |>
  dplyr::filter(CNTY_NAME %in% c("San Bernardino County"))
nrow(san_bernardino_bg)
```

```
[1] 1275
```

Check air quality and income values

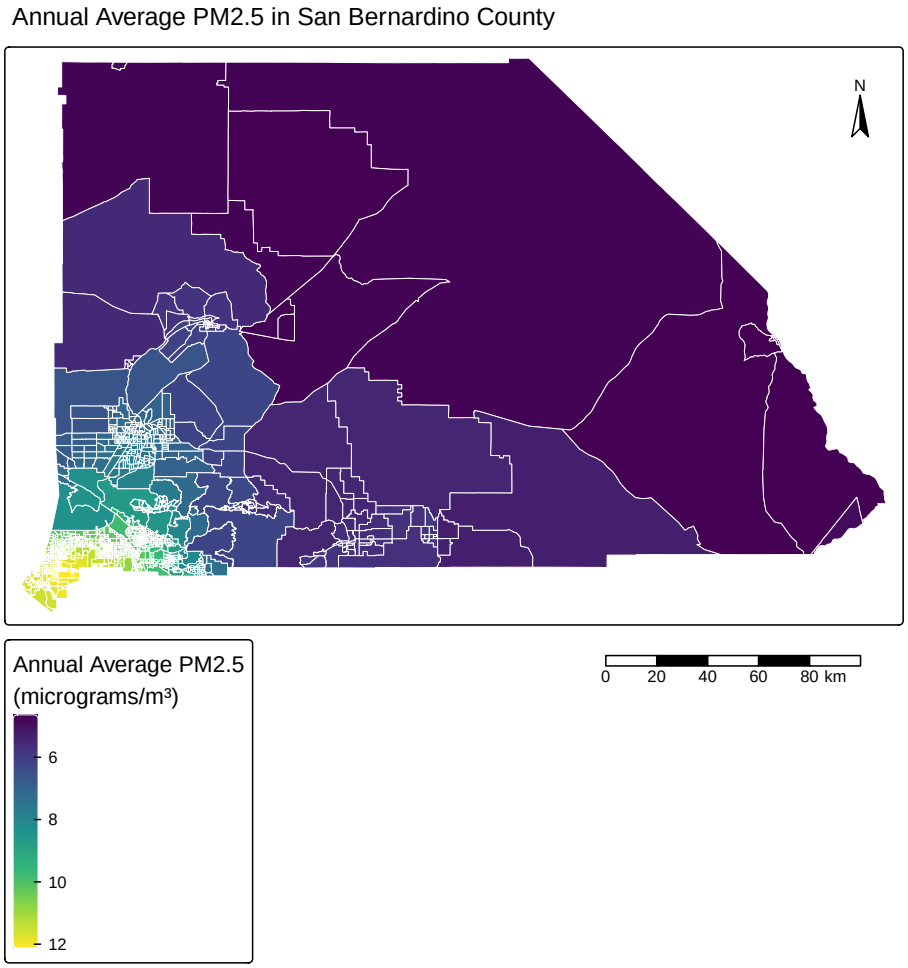
```r
san_bernardino_bg |>
  sf::st_drop_geometry() |>
  dplyr::select(ID, PM25, LOWINCPCT) |>
  summary()
```

```
      ID                 PM25           LOWINCPCT      
 Length:1275        Min.   : 4.608   Min.   :0.0000  
 Class :character   1st Qu.: 7.643   1st Qu.:0.1890  
 Mode  :character   Median : 9.967   Median :0.3260  
                    Mean   : 9.437   Mean   :0.3479  
                    3rd Qu.:10.978   3rd Qu.:0.4880  
                    Max.   :12.137   Max.   :1.0000  
```

Map 1, looking at PM 2.5 Levels in San Bernardino County, California

```r
# Map 1: Annual-average PM2.5 concentrations
tm_shape(san_bernardino_bg) +
  tm_polygons(
    fill = "PM25",
    lwd = 0.1,
    col = "white",
    fill.scale = tm_scale_continuous(
      values = "viridis"
    ),
    fill.legend = tm_legend(
      title = "Annual Average PM2.5\n(micrograms/m³)",
      orientation = "portrait",
      position = tm_pos_out(
```

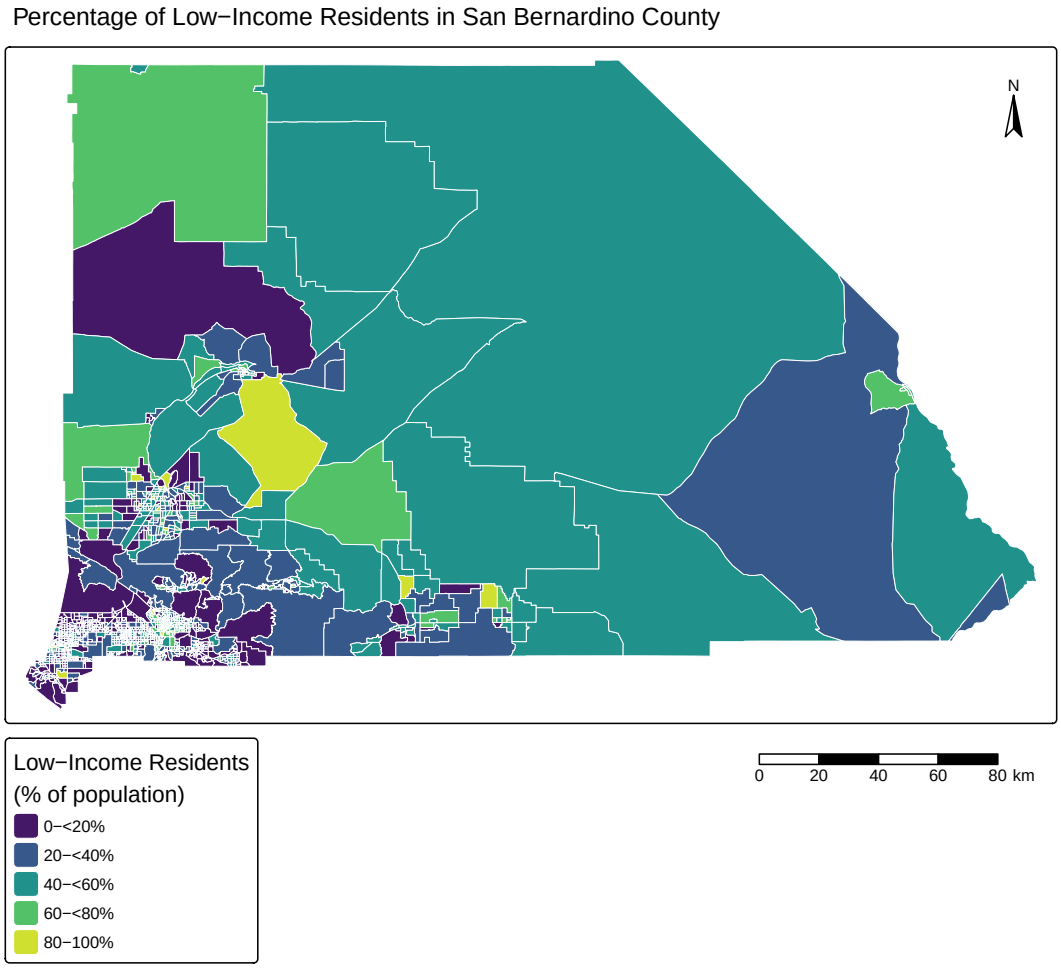
```
      "center", "bottom",
      pos.h = "left",
      pos.v = "top"
    )
  )
) +
tm_compass(
  size = 2,
  text.size = 0.7,
  position = tm_pos_in("right", "top")
) +
tm_scalebar(
  width = 15,
  text.size = 0.7,
  position = tm_pos_out(
    "center", "bottom",
    pos.h = "right", # horizontal position
    pos.v = "top" # vertical position
  )
) +
tm_title(
  "Annual Average PM2.5 in San Bernardino County",
  size = 1,
  position = tm_pos_out(
    "center", "top",
    pos.h = "left",# horizontal position
    pos.v = "bottom"# vertical position
  )
)
```



Map 2: Percentage of low-income residents

```
tm_shape(san_bernardino_bg) +
  tm_polygons(
    fill = "LOWINCPCT",
    lwd = 0.1,
    col = "white",
    fill.scale = tm_scale_intervals( #creating intervals for more readable map
      style = "fixed",
      breaks = c(0, 0.2, 0.4, 0.6, 0.8, 1),
```

```
    labels = c(
      "0-<20%",
      "20-<40%",
      "40-<60%",
      "60-<80%",
      "80-100%"),
    values = "viridis"),
  fill.legend = tm_legend(
    title = "Low-Income Residents\n(% of population)",
    orientation = "portrait",
    position = tm_pos_out(
      "center", "bottom",
      pos.h = "left", # horitzontal position
      pos.v = "top" )) ) + # vertical position
tm_compass(
  size = 2, text.size = 0.7,
  position = tm_pos_in("right", "top")) +
tm_scalebar(
  width = 15,
  text.size = 0.7,
  position = tm_pos_out(
    "center", "bottom",
    pos.h = "right",
    pos.v = "top")) +
tm_title( "Percentage of Low-Income Residents in San Bernardino County",
  size = 1,
  position = tm_pos_out(
    "center", "top",
    pos.h = "left", #horizontal position
    pos.v = "bottom"  # veritcal position
  ))
```



Map Results

The first map shows annual-average PM2.5 concentrations across census block groups in San Bernardino County. Concentrations are higher in the southwestern area of the county, while the northern and eastern areas have lower concentrations. The second map shows the percentage of lowm income residents in the county, and its pattern differs from the first map. In the southwestern area, there is a mix of block groups with both low and high percentages of low income residents. Some block groups farther north have higher percentages of low income residents and lower PM2.5 concentrations. Together, the maps suggest that higher PM2.5 concentrations do not consistently overlap with higher percentages of low-income residents across the county. It may be beneficial to examine the southwestern region more closely to

determine whether block groups with higher percentages of low-income residents also tend to have higher PM2.5 concentrations within that condensed region.